

Операционные системы

Продвинутое использование git

Махин Макар Михайлович

2026-09-03

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
mmmahin@mmmahin:~$  
mmmahin@mmmahin:~$ pnpm setup  
Appended new lines to /home/mmmahin/.bashrc  
  
Next configuration changes were made:  
export PNPM_HOME="/home/mmmahin/.local/share/pnpm"  
case ":$PATH:" in  
  *:$PNPM_HOME:*) ;;  
  *) export PATH="$PNPM_HOME:$PATH" ;;  
esac  
  
To start using pnpm, run:  
source /home/mmmahin/.bashrc  
mmmahin@mmmahin:~$ source ~/.bashrc  
mmmahin@mmmahin:~$
```

Рисунок 1: Node.js

```
mmmahin@mmmahin:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

WARN GET https://registry.npmjs.org/commitizen error (ECONNRESET). Will retry in 10 seconds. 2
retries left.
WARN Request took 29450ms: https://registry.npmjs.org/commitizen
WARN GET https://registry.npmjs.org/minimist error (ECONNRESET). Will retry in 10 seconds. 2 r
etries left.
WARN GET https://registry.npmjs.org/inquirer error (ECONNRESET). Will retry in 10 seconds. 2 r
etries left.
WARN Request took 29189ms: https://registry.npmjs.org/minimist
WARN Request took 29228ms: https://registry.npmjs.org/inquirer
WARN GET https://registry.npmjs.org/merge error (ECONNRESET). Will retry in 10 seconds. 2 retr
ies left.
WARN Request took 29391ms: https://registry.npmjs.org/merge
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/mmmahin/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Done in 1m 29.7s using pnpm v10.27.0
mmmahin@mmmahin:~$
```

Рисунок 2: установка commitizen

```
mmmahin@mmmahin:~$ pnpm add -g standard-changelog
WARN GET https://registry.npmjs.org/standard-changelog error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29476ms: https://registry.npmjs.org/standard-changelog
WARN GET https://registry.npmjs.org/@simple-lib%2Fchild-process-utils error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN GET https://registry.npmjs.org/@conventional-changelog%2Ftemplate error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN Request took 29163ms: https://registry.npmjs.org/@simple-lib%2Fchild-process-utils
WARN Request took 29174ms: https://registry.npmjs.org/@conventional-changelog%2Ftemplate
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +14
+++++++
Progress: resolved 181, reused 148, downloaded 14, added 14, done

/home/mmmahin/.local/share/pnpm/global/5:
+ standard-changelog 8.1.0

Done in 59.2s using pnpm v10.27.0
mmmahin@mmmahin:~$
```

Рисунок 3: установка standard-changelog

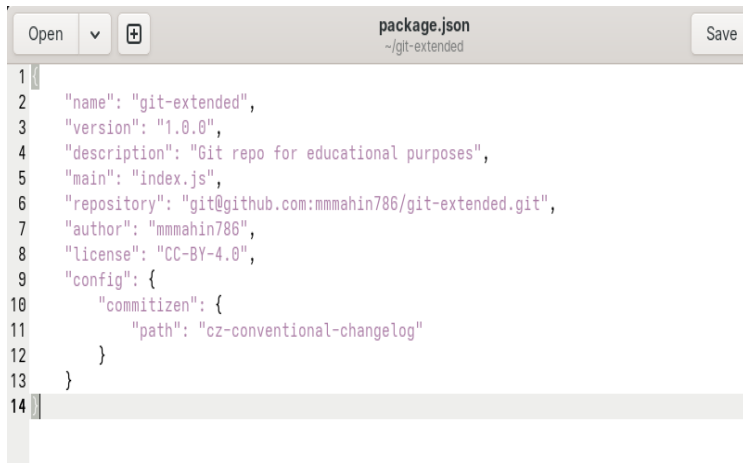


Рисунок 4: package.json

Использование commitizen

```
mmmahin@mmmahin:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore: Other changes that don't modify
src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master ac5dbd3] chore: test
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
mmmahin@mmmahin:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.11 KiB | 1.11 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mmmahin786/git-extended.git
 8b6b931..ac5dbd3  master -> master
mmmahin@mmmahin:~/git-extended$
```

Рисунок 5: Отправка

```
mmmahin@mmmahin:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/mmmahin/git-extended/.git/hooks]
mmmahin@mmmahin:~/git-extended$ git branch
* develop
  master
mmmahin@mmmahin:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/mmmahin786/git-extended/pull/new/develop
remote:
To github.com:mmmahin786/git-extended.git
* [new branch]      develop -> develop
mmmahin@mmmahin:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
mmmahin@mmmahin:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
mmmahin@mmmahin:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

mmmahin@mmmahin:~/git-extended$ standard-changelog --first-release
mmmahin@mmmahin:~/git-extended$ git add CHANGELOG.md
mmmahin@mmmahin:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 6fca9b1] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
mmmahin@mmmahin:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
mmmahin@mmmahin:~/git-extended$  
mmmahin@mmmahin:~/git-extended$ git push --all  
Enumerating objects: 6, done.  
Counting objects: 100% (6/6), done.  
Delta compression using up to 8 threads  
Compressing objects: 100% (4/4), done.  
Writing objects: 100% (5/5), 2.66 KiB | 2.66 MiB/s, done.  
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To github.com:mmmahin786/git-extended.git  
    ac5dbd3..90061ab  develop -> develop  
    ac5dbd3..16462f9  master -> master  
mmmahin@mmmahin:~/git-extended$ git push --tags  
Enumerating objects: 1, done.  
Counting objects: 100% (1/1), done.  
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.  
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To github.com:mmmahin786/git-extended.git  
* [new tag]          1.0.0 -> 1.0.0  
mmmahin@mmmahin:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md  
https://github.com/mmahin786/git-extended/releases/tag/v1.0.0  
mmmahin@mmmahin:~/git-extended$
```

Рисунок 8: Отправка

```
mmmahin@mmmahin:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

mmmahin@mmmahin:~/git-extended$ touch test2
mmmahin@mmmahin:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 90061ab).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

mmmahin@mmmahin:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

mmmahin@mmmahin:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.78 KiB | 2.78 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:mmmahin786/git-extended.git
   90061ab..91c0fe2  develop -> develop
   16462f9..b4df01d  master -> master
mmmahin@mmmahin:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 155 bytes | 155.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mmmahin786/git-extended.git
 * [new tag]         1.2.3 -> 1.2.3
mmmahin@mmmahin:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/mmahin786/git-extended/releases/tag/v1.2.3
mmmahin@mmmahin:~/git-extended$
```

Рисунок 10: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.